



# Functional Requirement Document

## EMERGENCY DESCENT function

Version **V01**

Date **09/03/2026**

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# 1 Executive summary

This document is the functional specification of the EMERGENCY DESCENT function. It provides “aircraft-level” requirements that will be the main input for systems specifications.

The purpose of the EMERGENCY DESCENT function is to have the aircraft descend to a safe altitude in the case of loss of cabin pressurisation.

# Contents

|          |                                |           |
|----------|--------------------------------|-----------|
| <b>1</b> | <b>Executive summary</b>       | <b>2</b>  |
| <b>2</b> | <b>Record of revisions</b>     | <b>5</b>  |
| <b>3</b> | <b>General</b>                 | <b>6</b>  |
| 3.1      | Objective . . . . .            | 6         |
| 3.2      | Applicability . . . . .        | 6         |
| 3.3      | Reference documents . . . . .  | 6         |
| 3.4      | Acronyms . . . . .             | 6         |
| 3.5      | Definitions . . . . .          | 6         |
| <b>4</b> | <b>Function Description</b>    | <b>7</b>  |
| <b>5</b> | <b>Operational Scenarios</b>   | <b>8</b>  |
| 5.1      | Nominal scenario 1 . . . . .   | 8         |
| 5.2      | Nominal scenario 2 . . . . .   | 8         |
| 5.3      | Additional scenarios . . . . . | 9         |
| <b>6</b> | <b>Requirements</b>            | <b>10</b> |
| <b>7</b> | <b>Open items</b>              | <b>11</b> |
| <b>8</b> | <b>Appendices</b>              | <b>12</b> |

# List of Figures

## 2 Record of revisions

| Issue | DATE       | Reason for revision |
|-------|------------|---------------------|
| V01   | 09/03/2026 | First issue         |

## 3 General

### 3.1 Objective

This document is the functional specification for the EMERGENCY DESCENT function.

- Section 4 provides a high-level description of the EMERGENCY DESCENT function.
- Section 5 describes operational scenarios of the EMERGENCY DESCENT function.
- Section 6 contains "aircraft level" requirements for the EMERGENCY DESCENT function.

### 3.2 Applicability

[1] is the baseline for this functional specification. Thus this document applies to any aircraft described by [1].

### 3.3 Reference documents

[1] Baseline avionics description.

[2] Arinc 702a-5: Advanced flight management computer system. 2018.

### 3.4 Acronyms

|     |                       |
|-----|-----------------------|
| AP  | Autopilot             |
| ND  | Navigation Display    |
| FCU | Flight Control Unit   |
| IAS | Indicated Air Speed   |
| MSA | Minimum Safe Altitude |

### 3.5 Definitions

## 4 Function Description

The purpose of the EMERGENCY DESCENT function is to navigate the aircraft from its current altitude to 10 000 ft or the MSA in the current sector, following the fastest descent profile.

The EMERGENCY DESCENT function provides the following features:

- It allows the flight crew to activate the EMERGENCY DESCENT.
- Upon activation of the EMERGENCY DESCENT:

The aircraft begins descending to 10 000 ft at 250 kts IAS.

It allows the flight crew to select a desired altitude if different from 10 000 ft.

It allows the flight crew to set a desired IAS for the descent.

- Upon arriving to the selected altitude, the aircraft levels off.
- It allows the flight crew to deactivate the EMERGENCY DESCENT, in which case the aircraft levels off at its current altitude.

**Note:** The flight crew is responsible for extending the airbrakes to increase the vertical speed of the descent. They are also responsible for the retraction of the airbrakes when the aircraft levels off.

## 5 Operational Scenarios

In nominal scenario 1, it is an Emergency descent with structural damage.

In nominal scenario 2, it is an Emergency descent without structural damage.

### 5.1 Nominal scenario 1

**Initial conditions:** The aircraft is flying at a typical A320 cruise altitude, both pilots are in the cockpit, the autopilot is engaged. An explosive depressurization occurs resulting in structural damage to the aircraft.

**STEP 1 :** A visual and an audio alert inform the pilots of the situation. The pilots put on their oxygen mask.

**STEP 2 :** The system sets the altitude to 10 000 ft and the speed to 250 kts ; and sets the heading to Selected Mode. There is an audio announcement of these actions, and the information is displayed on the FCU. The SPD/MACH knob is set to SPD. Terrain mode is displayed on the ND. The actions will be active (Managed Mode) after a delay (few seconds). Meanwhile, pilots can still modify the instructions before they become active.

**STEP 3 :** the system alerts the air traffic controller of the situation (Emergency depressurization and Emergency descent) + target altitude and speed (+ vertical speed ?).

**STEP 4 :** pilots manage the systems and communications with ATC. They deploy the speedbrakes as needed.

### 5.2 Nominal scenario 2

**Initial conditions:** The aircraft is flying at a typical A320 cruise altitude, both pilots are in the cockpit, the autopilot is engaged. An explosive depressurization occurs without structural damage to the aircraft.

**STEP 1 :** A visual and audio alert informs the pilots of the situation. The pilots put on their oxygen mask.

**STEP 2 :** The system sets the altitude to 10 000 ft and the speed to 250 kts ; and sets the heading to Selected Mode. There is an audio announcement of these actions, and the information is displayed on the FCU. The SPD/MACH knob is set to SPD. Terrain mode is displayed on the ND. The actions will be active (Managed Mode) after a delay.

Meanwhile, pilots can still modify the instructions before they become active.

**STEP 3 :** pilots set speed to selected mode and increase it to VMO.

**STEP 4 :** the system alerts the air traffic controller of the situation (Emergency depressurization and Emergency descent) + target altitude and speed (+ vertical speed ?).

**STEP 5 :** pilots manage the systems and communications with ATC. They deploy the speed brakes as needed.

## 5.3 Additional scenarios

None

(ideas for other scenarios : during climb, descent, spurious activation, ...)

## 6 Requirements

- REQ-1** : The system shall enable the aircraft to reach an altitude of 10000 ft without any intervention of the pilot in less than *TBD* minutes.
- REQ-2** : The system shall detect the presence of structural damage and adapt the managed speed accordingly.
- REQ-3** : In case of structural damage, the system shall maintain the managed speed of the aircraft at 250 kts.
- REQ-4** : In case of the absence of any structural damage, the system shall maintain the managed speed of the aircraft at  $V_{MO}$ .
- REQ-5** : The system shall set the HDG to Selected Mode, the SPD/MACH to SPD and display TERR on the ND.
- REQ-6** : Pilots shall be visually and audibly informed of each action the system is performing.
- REQ-7** : Pilots shall be able to cancel any spurious activation of the system in less than *TBD* seconds and with *TBD* buttons only (i.e., must be easy to control).
- REQ-8** : Pilots shall be able to modify any parameter set by the system in less than *TBD* seconds and with *TBD* buttons only (i.e., must be easy to control).
- REQ-9** : The system shall inform the ATCO of the EMER DES state of the aircraft, and the target parameters.

## 7 Open items

## 8 Appendices